Intro to Gen AI

Course Outline

1.0 COURSE DETAILS			
Course Title	Intro to Gen AI		
Major	**B.Tech. CS & AI, B.Tech. CS & DS**		
Credits	**4**	**Domain with Course Code**	**CSA233**
Mode	**Regular**	**Period**	**Even Semester 2026**
Academic Year and Semester	**2025-2026 Even Semester**	**Category (Core/Audit)**	**Core**
Instructor		**Office Hours**	
LTP (Lecture-Tutorial-Practical)	**2-1-2**		

2.0 SUBJECT OVERVIEW	
This course offers an **introductory exploration of Artificial Intelligence (AI) and Generative AI**, designed to build conceptual clarity and practical intuition across the field's major areas. Students will learn how AI systems **perceive, reason, learn, and generate**, covering a broad spectrum from classical machine learning to GenAI Through a blend of theory, demonstrations, and hands-on labs, learners will develop the ability to analyze AI techniques, experiment with real-world tools, and understand the opportunities and challenges of contemporary AI.	
Facilitator's Vision:	• To help students **build a broad and foundational understanding of AI and GenAI**, connecting traditional principles with emerging applications. The course emphasizes conceptual breadth, critical thinking, and exploratory practice over mathematical rigor or model optimization, creating a launchpad for deeper learning in advanced AI courses.

Relevance for Students:	• Artificial Intelligence is now at the core of innovation in every domain business, healthcare, education, design, and software systems. By surveying both **discriminative and generative approaches**, students gain a panoramic view of how intelligent systems are built and used. • This course prepares learners for **advanced study, applied research, and industry roles** in AI development by introducing them to the essential ideas, workflows, and ethical considerations that define the field today.
Uniqueness of the program	• *The course uniquely* ***integrates classical AI foundations with hands-on exposure to generative systems****, balancing intuition, coding, and real-world applications. Students will explore machine learning workflows, neural models, large language models, prompt engineering, fine-tuning, retrieval-augmented generation (RAG), and intelligent agents.* • *Labs use tools like* ***Python, Scikit-learn, Hugging Face, LangChain, and Streamlit****, enabling students to move from basic ML experiments to building simple GenAI prototypes.*
Pre-Requisites:	• Basic Python Programming • Introductory understanding of Linear Algebra and Probability • Familiarity with Machine Learning fundamentals (recommended)
Intellectual Goals & Guiding Questions:	• What distinguishes AI, ML, and Generative AI from one another? • How do machines represent and learn from data? • What are effective strategies for prompting and fine-tuning generative models? • How can retrieval and tool use make LLMs more useful and reliable? • What ethical issues arise in the use of generative systems, and how can they be mitigated?

3.0 EDUCATIONAL OBJECTIVES	
Objectives	***What the program will include to meet the requirements of the program goals.*** By the end of this course, students will be able to: **Understand** • Core concepts of Artificial Intelligence, including search, learning, reasoning, and perception. • The distinctions and relationships between AI, Machine Learning, Deep Learning, and Generative AI.

	• Fundamental building blocks of neural models and large language models (LLMs), including attention and transformers at a conceptual level. **Apply** • Basic data preprocessing, feature engineering, and model evaluation techniques using Python and Scikit-learn. • Prompting and fine-tuning approaches to interact with and adapt generative models for simple text, image, or code tasks. • Common GenAI tools and frameworks such as Hugging Face, OpenAI API, and LangChain for practical experimentation. **Analyze** • Model outputs using evaluation metrics and interpretability techniques to assess correctness, bias, and reliability. • The role of embeddings, retrieval, and context in improving generative model performance. • The trade-offs between prompting, fine-tuning, and retrieval-augmented generation (RAG). **Evaluate** • Ethical, social, and technical implications of deploying AI and GenAI systems, including bias, misinformation, and transparency concerns. • The strengths and limitations of different AI paradigms (rule-based, discriminative, generative) for varied problem domains.

	Create ● End-to-end AI mini-projects such as classification pipelines, chatbots with memory, or RAG-based assistants using Python libraries. ● Simple prototype applications with user interfaces using tools like Streamlit or Gradio to demonstrate GenAI functionality.

4.0 LEARNING OUTCOMES		
Outcomes	*Identify the learning outcomes based on Skills, Knowledge, Attitudes and Values that will be worked upon during the course.*	
	Skills	1. Design and implement end-to-end **AI workflows** including data preprocessing, model training, and evaluation using Python and Scikit-learn. 2. Apply **prompt engineering techniques** (zero-shot, few-shot, chain-of-thought) to guide and refine outputs from large language models (LLMs). 3. Use modern AI/GenAI frameworks such as **Hugging Face, OpenAI API, and LangChain** to build simple generative applications. 4. Develop and deploy small-scale AI prototypes (e.g., chatbots, RAG systems, or text summarizers) using **Streamlit or Gradio** for user interaction.
	Knowledge	1. Explain the **core concepts and evolution** of Artificial Intelligence, Machine Learning, Deep Learning, and Generative AI. 2. Describe the **types of learning** (supervised, unsupervised, reinforcement) and their associated problem formulations.

		3. Understand the **high-level architecture and functioning of transformers and attention mechanisms** that power LLMs. 4. Identify the roles of **embeddings, retrieval, and context** in enabling reasoning and grounded generation in modern AI systems. 5. Recognize the **ethical, social, and technical considerations** in building responsible and transparent AI systems.
	Attitudes/Values	1. Develop **curiosity and openness** to explore emerging AI technologies and evolving GenAI paradigms. 2. Appreciate the **interdisciplinary nature of AI**, integrating data, algorithms, and human values in problem-solving. 3. Cultivate a **responsible and ethical mindset** toward bias mitigation, fairness, and reliability in AI system design. 4. Value **continuous learning and experimentation** as key traits for engaging with the fast-changing AI ecosystem.

5.0 PROGRAM OUTLINE

Map the learnings of each week of the course

Week	Content (Lectures)	Activity (Focus / Key Topics)	Assignments
1 and 2	Course Orientation & Foundations of AI Faculty & course introduction, pedagogy, expectations AI vs ML vs DL; types of learning (supervised, unsupervised, reinforcement) Classification, regression, clustering tasks	Python/Numpy/Pandas refresher Setting up Jupyter environment Exploratory data analysis on toy dataset Build regression and classification models in Scikit-learn Compute evaluation metrics (MSE, accuracy, precision, recall, F1, ROC)	Short reflection: “Where do we see AI today?” Mini exercise: Data cleaning in Pandas Notebook submission: Implement regression and classification on a Kaggle dataset

	Classical ML Foundations Data preprocessing: missing values, scaling, encoding Linear Regression and Logistic Regression concepts & assumptions		
3 and 4	Model Evaluation and Decision Trees Evaluation metrics, bias–variance tradeoff Decision Trees: hyperparameters, assumptions, limitations Neural Networks Basics Introduction to neural networks and perceptron Multilayer perceptron and backpropagation (conceptual)	Train and visualize DecisionTreeClassifier using plot_tree Build ML pipeline: data cleaning, model training, evaluation Implement perceptron from scratch Build a simple MLP using PyTorch	Practical assignment: Design an ML pipeline for a small dataset Coding lab: Implement MLP on a small classification dataset
5 and 6	Introduction to Generative AI What is Generative AI? Key terminology, scope (text, image, code, audio) Discriminative vs generative models Transformer architecture (conceptual) and self-attention overview Responsible AI: bias, misinformation, fairness Prompt Engineering and Fine-Tuning Prompt design principles (zero-shot, few-shot, chain-of-thought)	Demo: Text generation (GPT/BERT), image generation (DALL·E/Stable Diffusion) Visualize attention maps using BertViz Simple text generation using Hugging Face pipeline Experiment with prompts in OpenAI Playground or API Fine-tuning walkthrough using Hugging Face TRL Compare outputs of base vs tuned model	Short essay: "How generative models differ from predictive models?" Notebook: Prompt variations for summarization/Q&A tasks

	Evaluating and refining LLM outputs Fine-tuning vs prompting, overview of PEFT methods		
7 and 8	LLM Tools and APIs Hugging Face Hub, Transformers, Pipelines Commercial APIs (OpenAI, Google AI): authentication, rate limits Introduction to embeddings and their role in retrieval Memory, Context, and Retrieval-Augmented Generation (RAG) Managing state and conversation history in chatbots RAG: grounding models with external knowledge Vector databases overview (Chroma, FAISS)	Use pipelines for text generation, classification, and summarization Generate embeddings for sentences Run inference using OpenAI API Implement multi-turn chatbot context management Build simple RAG pipeline (embed, retrieve, prompt)	Assignment: Create a Python script that uses an LLM API for Q&A Project Milestone 1: Submit chatbot or RAG prototype
9 and 10	Rapid Prototyping and Agentic AI Building GenAI apps using Streamlit/Gradio Agentic AI overview (LLM as brain, tools, memory, planning) The ReAct (Reason + Act) framework Advanced Agents and Automation Beyond simple agents: LangGraph	Build UI for chatbot or RAG system using Streamlit/Gradio Implement a simple LangChain agent with tools (calculator/web search) Create AI workflow in n8n triggered by webhook Test AI Agent node integration with different apps like slack etc	Project Milestone 2: Streamlit-based functional GenAI app Final Project Submission & Demo: Build a connected AI workflow using LLMs

	concepts (nodes, edges, state) Low-code AI workflows with n8n Connecting LLMs to APIs, triggers, and services		

6.0 EVALUATION CRITERIA *Map the percentage weightage to each category.*		
Function	**% Weightage**	**Notes**
Attendance/Class Participation	**5%**	
In Class/Post Class Assignments	**5%**	
Project	**20%**	
Contests	**10%**	
Mid Semester	**20%**	
End Semester	**40 %**	

7.0 REQUIRED READINGS AND REFERENCES List of required and suggested material. Can mention week-wise or topic-wise. Can also mention how to access readings.	
Readings	1. **(Primary Text)** Russell, Stuart, and Peter Norvig. *Artificial Intelligence: A Modern Approach*. 4th ed., Pearson, 2021. *(Comprehensive introduction to foundational AI concepts, search, reasoning, and learning.)* 2. **(Reference)** Géron, Aurélien. *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*. 3rd ed., O'Reilly Media, 2023. *(Excellent practical companion for supervised/unsupervised learning and neural network basics.)* 3. **(Reference)** Chollet, François. *Deep Learning with Python*. 2nd ed., Manning Publications, 2021. *(Clear introduction to deep learning and neural architectures leading up to transformers.)* 4. **(Reference)** Tunstall, Lewis, Leandro von Werra, and Thomas Wolf. *Natural Language Processing with Transformers: Building Language Applications with Hugging Face*. O'Reilly Media, 2022. *(Accessible and hands-on guide to transformers, fine-tuning, and modern GenAI workflows.)*
Videos	1. **AI For Everyone** – Andrew Ng (Coursera, DeepLearning.AI) *Conceptual and ethical grounding in AI applications.* 2. **Generative AI with LLMs** – Andrew Ng & Isa Fulford (DeepLearning.AI Short Course) *Hands-on introduction to prompt engineering, fine-tuning, and retrieval-augmented generation.* 3. **MIT OpenCourseWare: 6.034 – Artificial Intelligence** (MIT) *Classical AI foundations: search, logic, planning, and learning.* 4. **NPTEL Course:** *Introduction to Artificial Intelligence* by Prof. Mausam (IIT Delhi) *Strong conceptual grounding for Indian academic context.*
Others	1. **Course Management System (LMS/CMS):** For lecture slides, lab notebooks, announcements, and assignment submissions.

	2. **Discussion Platform:** slack/mail 3. **Kaggle / Google Colab Integration:** For lab-based experiments, data exploration, and building AI/GenAI prototypes. 4. **APIs & Tools:** Hugging Face Hub, OpenAI API, LangChain, Streamlit, Gradio, and n8n for building generative applications.

8.0 PROGRAM AUTHORIZATION

(SIGNOFF)

The undersigned acknowledge they have reviewed the course outline. Changes to this course outline will be coordinated with and approved by the undersigned or their designated representatives.

Program Head	Signature: ____________ Date: ______ Print Name: ____________ Title: ____________ Role: ____________
	Signature: ____________ Date: ______ Print Name: ____________ Title: ____________ Role: ____________
OAA Office	Signature: ____________ Date: ______ Print Name: ____________ Title: ____________ Role: ____________
VC Office	Signature: ____________ Date: ______

	Print Name: ______________________ Title: ______________________ Role: ______________________
	Signature: ______________________ Date: __________ Print Name: ______________________ Title: ______________________ Role: ______________________
Course Owner	Signature: ______________________ Date: __________ Print Name: ______________________ Title: ______________________ Role: ______________________
	Signature: ______________________ Date: __________ Print Name: ______________________ Title: ______________________ Role: ______________________
	Signature: ______________________ Date: __________ Print Name: ______________________

	Title: ______________________ Role: ______________________